# The map on a fingertip

Fingerprints

Name _____ Date _____



Morning light crossed the table, and a faint swirl appeared on a drinking glass. Little Wonderer tilted the magnifier. The mark was not one solid smudge. It held many curved lines. Big wondered aloud, “What part of a fingertip could leave a map like that?”



A close look at a fingertip revealed raised lines with lower spaces between them. The raised lines were friction ridges; the lower spaces were grooves. Only the ridge tops had touched the glass, leaving a flat impression of the fingertip's textured surface.



Big placed three field cards on the table. One pattern curved back toward the side: a loop. One circled a center: a whorl. One rose like a hill and crossed the card: an arch. These families described the broad flow, but they did not tell the whole story.

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|---|---|
| **fingerprint** | an impression that shows the friction ridges of a fingertip |
| **friction ridge** | one raised line on the skin of a fingertip |
| **groove** | a lower space between two raised ridges |
| **impression** | a mark made when one surface presses another |
| **loop** | a ridge flow that curves back toward the side it came from |
| **whorl** | a ridge flow that circles around a center |
| **arch** | a ridge flow that rises and crosses from side to side |
| **ridge ending** | a place where one ridge stops |
| **bifurcation** | a place where one ridge splits into two paths |
| **clarity** | how clearly the details can be seen |

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Two cards both held loops. At first they seemed alike. Then Little found a ridge that ended on one card and continued on the other. Another ridge split into two paths at a different place. The tiny endings and splits turned each broad loop into a more detailed map.



Little touched one fingertip to a washable stamp pad, then pressed paper once. Ink rode on the raised ridge tops. The lower grooves stayed pale. The paper held an impression: dark ridge paths with light spaces between them. After one careful look, the print stayed at home and hands were washed.



A second press looked lighter and a little smudged, but the same turns, endings, and splits still lined up. Pressure had changed the impression's clarity, not the ridge map beneath it. Little set a different fingertip print beside the pair. Its broad flow and tiny details made a new map to explore.

1. How do raised ridges and lower grooves make the light and dark lines in an impression?
2. Why are loop, whorl, and arch useful clues but not the whole fingerprint map?
3. What evidence could show that a light print and a dark print came from the same fingertip?

## Your turn

Write a compare sentence about two ridge maps.

Example: Both prints are loops, but this ridge ends sooner.